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Comparison of the combustion properties of methane, hydrogen, and propane.

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1. Introduction

Methane (CH_4), hydrogen (H_2), and propane (C_3H_8) are three of the most widely used gaseous fuels in industrial, residential, and emerging energy applications. While all three support stable combustion under appropriate conditions, their thermochemical behavior differs in ways that matter greatly for engineering design, safety, and environmental impact. Among the many parameters used to characterize a fuel's combustion behavior, flame temperature and combustion product composition are two of the most fundamental.

Flame temperature describes how hot a flame becomes under a given set of conditions — most rigorously expressed as the adiabatic flame temperature, which assumes all heat released by combustion remains within the combustion products with no losses to the surroundings. This quantity directly determines how much thermal energy is available for a given application, whether that is heating a furnace, driving a gas turbine, or powering a domestic burner. Flame temperature also governs secondary phenomena of considerable practical importance: at higher temperatures, thermal decomposition and dissociation of product species become significant, and the formation of nitrogen oxides (NO_x) through the thermal NO_x pathway accelerates sharply. A fuel's flame temperature therefore sits at the intersection of thermal performance and emissions behavior.

Combustion product composition refers to the chemical species present in the exhaust gases once the fuel has reacted with an oxidizer. In idealized complete combustion, the products are determined straightforwardly by the elemental composition of the fuel — carbon yields CO_2, hydrogen yields H_2O, and so on. In practice, however, real flames operate under non-ideal conditions: local fuel-to-air ratio variations, finite reaction rates, and high temperatures all give rise to a broader spectrum of species, including carbon monoxide (CO), unburned hydrocarbons, and various nitrogen oxides. Understanding product composition is essential for assessing a fuel's carbon footprint, designing exhaust treatment systems, ensuring combustion completeness, and meeting regulatory emission standards.

Together, flame temperature and product composition provide a concise yet powerful lens through which the suitability of a fuel can be evaluated for any given application — making their comparison across methane, hydrogen, and propane both practically relevant and scientifically instructive. Understanding these differences lays the groundwork for a rigorous comparison across further combustion parameters, including flammability limits, laminar burning velocity, and specific energy content.

2. Methodology

The purpose of these calculations is to compare the adiabatic flame temperature and the composition of the combustion products of selected gaseous fuels: methane (CH_4), hydrogen (H_2), and propane (C_3H_8) in a mixture with air.

The following initial conditions were adopted:

T_0 = 288,15 K
p_0 = 1 atm = 101 325 Pa
φ = 1

The combustion process is described using following stoichiometric reactions:

Methane (CH_4):

$$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$$

Hydrogen (H_2):

$$H_2 + 0{,}5\ O_2 \rightarrow H_2O$$

Propane (C_3H_8):

$$C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$$

In the actual combustion process many intermediate reactions occur; therefore, instead of a single reaction equation, the *gri30.yaml* kinetic mechanism available in the Cantera module was used. This mechanism describes a system of multiple chemical species and elementary reactions.

The flame temperature was calculated based on the following assumptions:

- adiabatic combustion,
- constant pressure,
- no heat loss to the surroundings,
- perfect fuel-air mixture,
- the chemical balance of the products.

Therefore implying:

$$H_{reactants} = H_{products}$$

Cantera determines the flame temperature by iteratively solving for chemical equilibrium:

$$Hp = const.$$

where:

- H – enthalpy,
- p – pressure.

Once chemical equilibrium has been reached, the composition of the post-reaction mixture is determined. The molar fraction of a given component is calculated as:

$$X_i = \frac{n_i}{\sum n_i},$$

where:

- X_i - molar fraction of a component
- n_i - number of moles of a component.

The results are presented as percentages.

3. Code

The following report is based around computer-based calculations using python programming language with following modules: Cantera, NumPy, and Matplotlib (Pyplot submodule). Code can be accessed via GitHub platform:

https://github.com/MilkyDrama/MKwS

4. Results

Methane (CH_4):

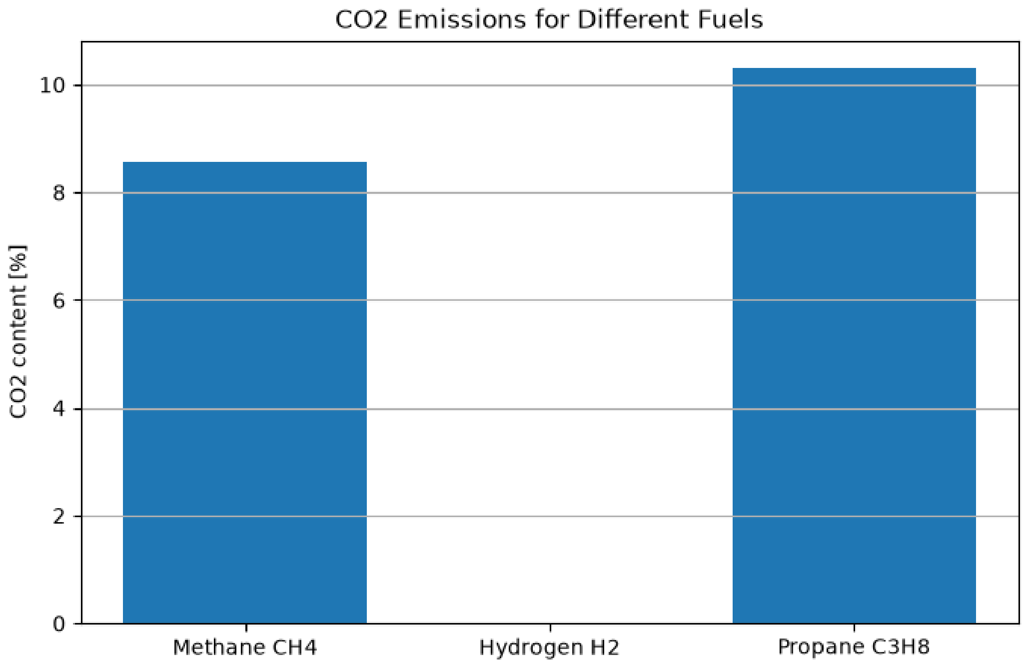
Flame Temperature: 2219,7 K
CO_2: 8,56 %
H_2O: 18,36 %
O_2: 0,45 %

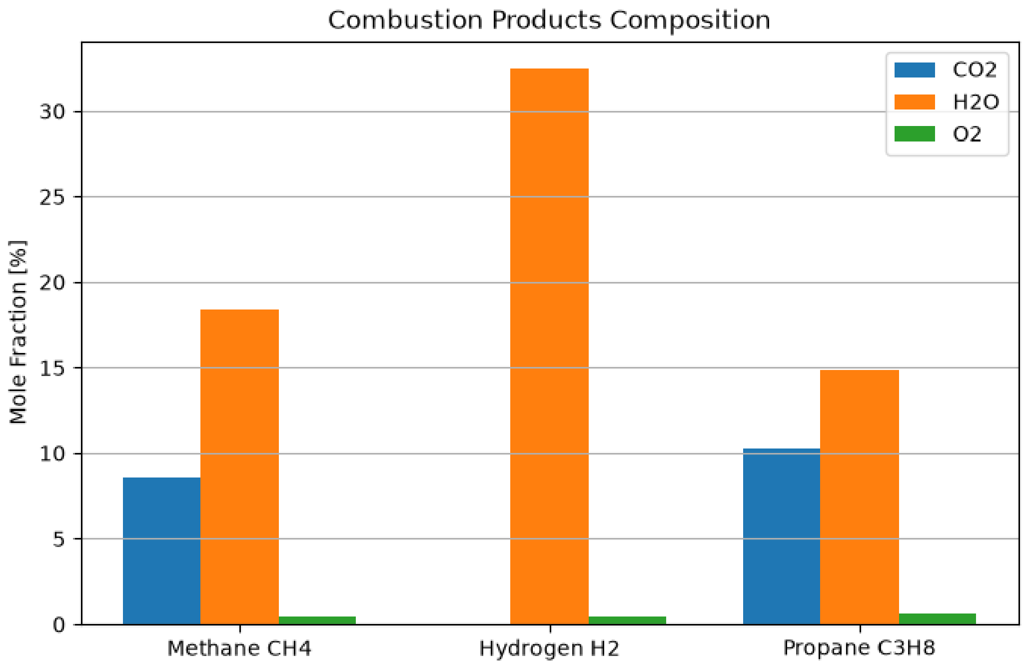
Hydrogen (H_2):

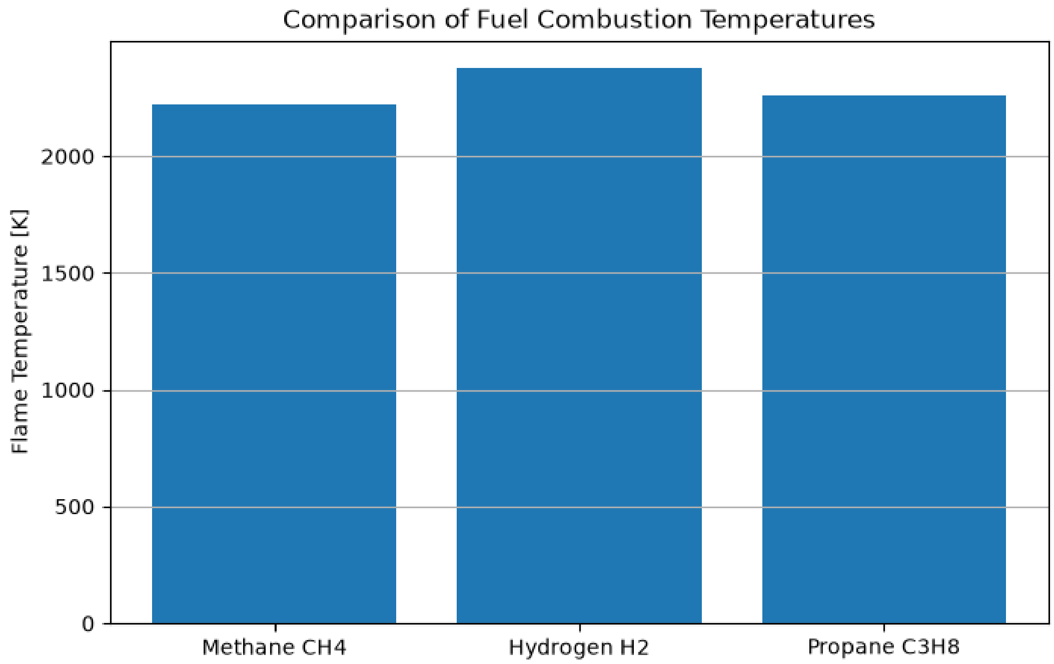
Flame Temperature: 2374,7 K
CO_2: 0,00 %
H_2O: 32,45 %
O_2: 0,47 %

Propane (C_3H_8):

Flame Temperature: 2261,1 K
CO_2: 10,30 %
H_2O: 14,86 %
O_2: 0,58 %


CO2 Emissions for Different Fuels
CO2 content [%]
0
2
4
6
8
10
Methane CH4
Hydrogen H2
Propane C3H8


Combustion Products Composition
Mole Fraction [%]
0
5
10
15
20
25
30
CO2
H2O
O2
Methane CH4
Hydrogen H2
Propane C3H8



5. Summary

The results show that hydrogen combustion produces the highest flame temperature. This is due to the high reactivity of this fuel and the high rate of the chemical reactions involved. The combustion of methane and propane results in slightly lower temperatures, but these values remain similar due to the comparable nature of the combustion process under stoichiometric conditions.

An analysis of the exhaust gas composition showed that the main products of hydrocarbon fuel combustion are carbon dioxide and water vapor. In the case of methane and propane, the presence of CO_2 results directly from the carbon content in the fuel molecule. Propane generates more CO_2 than methane because it contains more carbon atoms per fuel molecule. Methane, on the other hand, has lower CO_2 emissions per unit of energy compared to heavier hydrocarbons. No CO_2 is produced when hydrogen is burned because the fuel contains no carbon atoms. The main product of hydrogen combustion is water vapor, which points to hydrogen's potential advantage as a fuel in terms of reducing greenhouse gas emissions. The small amount of oxygen in combustion products results from the attainment of chemical equilibrium—even for a stoichiometric mixture, some oxygen remains in the system after the reaction is complete.